Code -Layout, Readability and Reusability

Date	30 June, 2026
Team ID	SWTID-2026-8797
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the Credit Card Approval Prediction project code in terms of structure, readability, maintainability, and reusability. The project is developed using Python, Flask, and Machine Learning techniques. Proper coding standards, modular programming, and meaningful documentation make the application easy to understand, maintain, and extend for future enhancements.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform indentation and spacing maintained throughout the Python code.	Yes	Improves readability and debugging.
2	Proper File Structure	Files are organized into templates, static, model, dataset, and application folders.	Yes	Follows Flask project structure
3	Meaningful Variable Names	Variables such as income , credit_score, employment_years, and prediction clearly indicate their purpose.	Yes	Easy to understand
4	Function / Method Names	Functions like predict(), load_model(), and preprocess_data() are descriptively named.	Yes	Enhances maintainability.
5	Code comments	Important sections contain comments explaining preprocessing, model loading, and prediction logic.	Yes	Helps future developers.
6	Modular Design	Data preprocessing, model training, evaluation, and web application are separated into different modules.	Yes	Highly reusable.

7	No Redundant Code	Duplicate code has been minim ized by using reusable functions.	Yes	Cleaner implementation.
8	Error H andling	Invalid user inputs and prediction erro rs ar e handled using exception handli ng.	Yes	Prevents applicati on crashes.

Reus a ble Compo nents / Mo dules:

S.No	Component / Module Name	Lan gu age / Techn olo gy	Where Used	Reusab il ity Level (High / Mediu m / Low)
1	Flask Web Applic ation	Flask, HTML, CS S	User Interfa ce	High
2	Feature Encoding Module	Scikit -learn	Dataset Preparati on	High
3	Machine Le arning Model	Random F orest / Logist ic Regression	Prediction System	High

Aspect	Ratin g(1-5)	Comments
Code Layout & Structure	5	Well-org anized proj ect di rectory following Flask architecture.
Readability	5	Meaningful variabl e nam es, proper formatti ng, and comm ent s im prove readabil it y.
Reusabilit y	5	Modular functions and re usable ML components allow future enhancements.
Documentation / Comm ents	5	Adequate comments explain i mportant sections of the code.
Overall S cor e	5/5	Excellent code qu ality wit h proper structure, r eadabil it y, and maintainabil it y.

Ov erall Code Quality Assess ment: